Bassel Ashraf El-Bahnasy

COMPUTER ENGINEER · DATA SCIENCE & CYBERSECURITY

+201120903622 | basselashraf@hotmail.com | Hadayek October, Giza, Egypt
LinkedIn | GitHub

EDUCATION

Bachelor's Degree in Computer Engineering
King Salman International University · GPA: 3.513

EXPERIENCE

Internship Trainee — Commercial International Bank (CIB)

- Analyzed ESG frameworks, climate risk, and corporate credit risk to understand sustainable lending practices.
- Explored applications of AI, blockchain, and data analytics in financial risk modeling and enterprise resilience, while developing presentation and customer-centric skills.

Data Engineer Intern — Digital Egypt Pioneers Initiative

- Developed and managed data pipelines using Azure Data Factory, with ETL workflows loading into Azure SQL Databases.
- Applied machine learning algorithms for data insights and predictive modeling.
- Led a team of peers to complete end-to-end data engineering projects, delivering results on schedule; earned a leadership certification for team management.

Freelance AI Data Trainer & Annotator — Outlier, Alignerr, Telus International, Upwork, Silencio AI, RWS

- Contributed to large-scale AI model training and evaluation as a freelance data annotator across multiple platforms (Outlier, Alignerr, Telus International), reviewing model outputs and labeling data to support machine learning fine-tuning since April 2023.
- Delivered specialized data-collection projects via Upwork (audio replay data collection) and Silencio AI, maintaining data quality and accuracy standards for client deliverables.

PROJECTS

Phoenix — Autonomous Fire-Detection & Suppression Robot
King Salman International University · Python, ROS 2, OpenCV, MQTT, Raspberry Pi 4

- Designed a split-node robotics architecture separating a vision node, a simulation environment, and physical edge hardware for autonomous flame detection and suppression.
- Built a computer vision pipeline using ArUco marker-based pose estimation and a solvePnP transformation matrix to calculate real-time fire distance with a safe stopping offset.
- Implemented ROS 2 navigation and mapping using Nav2 and SLAM Toolbox, validated in Gazebo simulation before hardware deployment.
- Controlled BTS7960 motor drivers with smooth speed control and velocity interpolation, and drove dual MG996R servos via pigpio DMA-backed hardware PWM for jitter-free motion.
- Migrated the MQTT communication layer from HiveMQ Cloud to a self-hosted Mosquitto broker, cutting latency and removing dependence on internet connectivity for fire/human-detection alerts and fail-safe triggers.
- Built a web-based command center for live monitoring and control of the robot.

GitHub: github.com/Bassel1000/phoenix_robot

TRAINING & WORKSHOPS

Embedded Systems Workshop — 60-hour program covering C/Embedded C, computer architecture, microcontroller interfacing, and RTOS.

Huawei Data Communication Training — Two-week HCIA-equivalent training covering TCP/IP, routing, switching, VLANs, and subnetting (Huawei eNSP).

Siemens PLC Training — Two courses in industrial automation: Siemens S7-1200 PLC programming via TIA Portal, ladder logic optimization.

Electronics Factory Training — Two-week program in PCB design, Surface Mount Technology (SMT), soldering, and QC for consumer electronics.

SPECIALIZATIONS

Google Advanced Data Analytics Specialization

7 courses covering statistical analysis, machine learning, and predictive modeling, including Foundations of Data Science, Regression Analysis, and a Capstone project. Gained expertise in Python, SQL, and data visualization for data-driven decision-making.

Google Data Analytics Professional Certificate

8 courses covering data cleaning, exploration, SQL, R programming, and dashboard creation, culminating in a case-study capstone.

COURSES & CERTIFICATIONS

Robotic Mapping and Trajectory Generation — *University of Colorado Boulder* · April 2026

Network Security — *ISC2* · April 2026

Claude Code in Action — *Anthropic* · April 2026

AI Fluency Framework & Foundations — *Anthropic* · April 2026

Hardware Security — *University of Maryland* · November 2025

IoT Devices — *University of Illinois Urbana-Champaign* · November 2025

Responsible Innovation and Trustworthy AI — *SAS* · July 2025

Artificial Intelligence Fundamentals with Capstone Project — *IBM* · May 2025

Parallel Programming — *EPFL* · April 2025

Ethical Hacking Fundamentals — *LearnKartS* · April 2025

Fundamentals of Accelerated Computing with CUDA Python — *Nvidia* · February 2025

Modern Robotics, Course 3: Robot Dynamics — *Northwestern University* · December 2024

Digital Signal Processing 1: Basic Concepts and Algorithms — *EPFL* · December 2024

Cryptography — *University of Maryland* · November 2024

Modern Robotics, Course 2: Robot Kinematics — *Northwestern University* · November 2024

SKILLS

Data Science & Analytics:
Machine Learning, Predictive & Regression Modeling, Statistical Analysis, Data Cleaning, Aggregation, and Visualization.

Cybersecurity: Network Security, Threat Detection, Network Vulnerabilities, Data Privacy/Confidentiality, Ethical Hacking Fundamentals.

Programming Languages: Python (incl. CUDA), C++, Java, SQL, R.

Robotics & Embedded: ROS 2, Gazebo, SLAM/Nav2, OpenCV, pigpio/GPIO, Embedded C, RTOS.

Tools & Infrastructure: Tableau, Jupyter Notebook, Kaggle, Microsoft Azure, Microsoft SQL Server, Visual Studio, VS Code, Siemens TIA Portal.

Languages: Arabic (Native), English (C1 – Advanced).

COMPETITIONS

- ICPC — International Collegiate Programming Contest
- Egypt AI & IoT Challenge 2026 — Top 10, National Graduation Projects